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Page: 1/31

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TRADE AND ENVIRONMENTAL SUSTAINABILITY STRUCTURED DISCUSSIONS (TESSD)

STATEMENT BY THE TESSD CO-CONVENORS

Addendum

This addendum includes *TESSD: Insights and Outcomes from Five Years of Work – A Co Convenors' Report*, accompanying the Statement by the TESSD Co-Convenors circulated in document WT/MIN(26)/22.

TESSD: INSIGHTS AND OUTCOMES FROM FIVE YEARS OF WORK – A CO-CONVENORS' REPORT

Contents

MESSAGE FROM THE CO-CONVENORS

Over the past decade, it has become increasingly clear that environmental challenges facing our planet – from climate change and biodiversity loss to pollution and resource depletion – are deeply intertwined with the ways we trade, produce, and innovate. In response to this growing awareness, a group of WTO Members have worked together to progressively create new spaces within the multilateral trading system for trade and environmental communities to engage constructively, learn from one another, and chart a shared path forward.

This journey began in 2018 with the creation of the Friends Advancing Sustainable Trade (FAST) group by Canada and Costa Rica. Launched in Davos by nine Members[1], FAST was founded on a simple yet ambitious premise: that trade can and should be a powerful enabler of the Sustainable Development Goals. It sought to stimulate new thinking at the intersection of trade and the environment, bridge gaps between policy communities, and encourage engagement and leadership among international organizations, business, academia, and civil society.

Building on this momentum, a group of WTO Members established the Trade and Environmental Sustainability Structured Discussions (TESSD) in 2020, as a broader, Member-driven platform for structured, transparent, and inclusive dialogue. Since then, TESSD has grown significantly, with Members from all geographies, levels of development, with specific challenges and interests, allowing it to develop a solid, and representative, technical foundation through thematic discussions, expert engagement, and sustained stakeholder participation. These discussions have deepened the understanding of key issues such as environmental goods and services, circular economy and circularity, environmental effects and trade impacts of relevant subsidies, value chain transparency, and capacity-building needs, particularly for developing and least developed Members.

The Ministerial Statement issued for the 12th WTO Ministerial Conference (MC12) held in June 2022 reaffirmed Members' commitment to this agenda and underscored the importance of dialogue and information sharing on trade and environmental sustainability. It confirmed that this work is integral to the relevance, credibility, and the future of the WTO, and highlighted the need to identify concrete actions to support environmental sustainability.

This document traces the evolution of TESSD from its early origins to its current role as a platform for innovative dialogue, learning, and collaboration. It illustrates how openness, inclusiveness, and sustained technical engagement can build shared understanding among Members. Above all, it reflects a growing recognition that trade can contribute to environmental sustainability, innovation, and the development of pathways that are both more sustainable and equitable.

We hope this publication serves not only as a record of progress, but also as an invitation to continue building bridges, deepening ambition, and translating dialogue into meaningful action for present and future generations.

Nadia Theodore Ambassador, Permanent Representative of Canada to the WTO	Ronald Saborío Ambassador, Permanent Representative of Costa Rica to the WTO

[1] Australia, Canada, Chinese Taipei, Costa Rica, the European Union, Mexico, New Zealand, Norway, and Switzerland.

1 EVOLUTION OF TESSD

1.1. In the wake of the COVID pandemic and amid growing concerns regarding pollution, biodiversity loss and climate change, 50 WTO Members launched the Trade and Environmental Sustainability Structured Discussions in November 2020, during the WTO's inaugural Trade and Environment Week.

> **What is TESSD?**
> TESSD is an initiative open to all WTO Members and invited stakeholders, designed to complement and advance WTO discussions on trade and environmental sustainability. It serves as an incubator and "test lab" for new ideas, as well as a forum to examine how trade and trade policy can support environmental objectives. In doing so, TESSD seeks to identify concrete actions that participating Members may pursue individually or collectively to expand opportunities for environmentally sustainable trade.

1.2. Recognizing that international trade and trade policy play a pivotal role in enabling the transition towards a climate neutral, more resource-efficient, and more circular global economy, the proponents called for environmental sustainability be established as a guiding principle in the broader WTO reform efforts. To this end, they initiated discussions to identify common topics of interest, with the aim of complementing the multilateral work taking place in other WTO bodies, particularly the Committee on Trade and Environment (CTE).

1.3. By December 2021, participation in TESSD had expanded to 71 Members. At the WTO's 12th Ministerial Conference (MC12), TESSD trade ministers issued a Ministerial Statement underscoring that international trade and trade policy can and must support environmental and climate goals, while promoting environmentally sustainable patterns of production and consumption. The Statement also highlighted the importance of a just transition and reaffirmed Members' commitments to advancing the Sustainable Development Goals (SDGs).

Graphic 1: TESSD evolution timeline

1.4. To structure their work and identify concrete actions that Members could pursue individually or collectively to advance environmentally sustainable trade, including through voluntary actions and partnerships, participants established four Working Groups (WGs) on: (i) Trade-related Climate Measures (TrCMs); (ii) Environmental Goods and Services (EGS); (iii) Circular Economy – Circularity; and (iv) Subsidies.

1.5. Coordinated by the Co-Convenors, Canada and Costa Rica, the WGs are supported by facilitators who help shape the meeting agendas, solicit expert contributions, and prepare guiding questions to focus the discussions (see boxes describing each WG). In December 2022, a high-level stocktaking event reviewed progress, identified priorities for continued work, and affirmed TESSD's potential to positively contribute to global trade and environmental sustainability discussions.

> **TESSD's 79 Co-sponsors**
> Albania; Argentina; Australia; Austria; Bahrain, Kingdom of; Barbados; Belgium; Brazil; Bulgaria; Cabo Verde; Canada; Chad; Chile; China; Colombia; Costa Rica; Croatia; Cyprus; Czech Republic; Denmark; Ecuador; Estonia; European Union; Fiji; Finland; France; the Gambia; Germany; Greece; Honduras; Hong Kong, China; Hungary; Iceland; Ireland; Israel; Italy; Japan; Kazakhstan; Korea, Republic of; Latvia; Liechtenstein; Lithuania; Luxembourg; Macao, China; Maldives; Malta; Mexico; Moldova, Republic of; Montenegro; Netherlands; New Zealand; North Macedonia; Norway; Panama; Peru; Philippines; Poland; Portugal; Romania; Russian Federation; Saudi Arabia, Kingdom of; Senegal; Singapore; Slovak Republic; Slovenia; Spain; Suriname; Sweden; Switzerland; Separate Customs Territory of Taiwan, Penghu, Kinmen and Matsu; Tajikistan; Thailand; Türkiye; Ukraine; United Arab Emirates; United Kingdom; United States; Uruguay; and Vanuatu.

1.6. TESSD is a pioneer initiative in its incorporation of stakeholder participation and contributions. Given the technical and crosscutting nature of trade and environment issues, inputs from experts across academia, civil society, non-government organizations (NGOs), intergovernmental organizations (IGOs), and the private sector have been essential in informing Members' work.[2] Delegations have also shared experiences and domestic practices, highlighting trade-related challenges, opportunities, and solutions.

Working Group on TrCMs

Facilitator: Switzerland
Overview: Members exchanged experiences and explored practical ways to enhance cooperation on the design and use of TrCMs in support of climate objectives. Discussions paid particular attention to developing country perspectives.

Working Group on Circular Economy – Circularity

Facilitators: Japan and Türkiye
Overview: Members examined trade-related aspects of circular economy across the full product lifecycle and shared policy experiences from a broad range of sectors. Discussions included consideration of development perspectives.

Working Group on EGS

Facilitators: The Philippines and the United Kingdom
Overview: Using an objective-based approach, and with particular attention to issues of interest to developing countries, Members explored how trade in EGS can contribute to climate mitigation and adaptation. Discussion addressed sector-specific issues related to the promotion and facilitation of trade in EGS.

Working Group on Subsidies

Facilitators: Israel and the Republic of Korea
Overview: Members examined both the potential environmental benefits and adverse impacts of subsidies, as well as their implications for trade. Discussions focused on sharing experiences and exploring ways to improve transparency, data availability, and understanding of subsidy practices.

1.7. Over the past five years, TESSD has implemented an intensive work programme. Through 25 sets of meetings, side events, and workshops, nearly 100 delegations and stakeholders delivered more than 160 presentations, most of which – together with informal meeting summaries – are publicly available online. Additional perspectives, experiences and positions were shared through subsequent discussions. The wealth of information generated underscores the deep interlinkages between trade, sustainable development, and environmental protection.[3]

1.8. The insights generated through this work are reflected in the package of substantive outcome documents developed by the four WGs and presented at the WTO's 13th Ministerial Conference (MC13) in Abu Dhabi in February 2024 (see below). The TESSD MC13 package included a Statement by the TESSD Co-convenors and an Updated Work Plan outlining the path forward in the four Working Groups. This work plan focusses on enhancing transparency, integrating development perspectives, and identifying best practices and policy opportunities, with the objective of delivering concrete outcomes by MC14.

MC13 Working Groups Outcome Documents

- Member Practices in the Development of Trade-related Climate Measures (TrCMs): Provides a compilation of Member practices in the development of trade-related climate measures, with focus on: (i) transparency and consultations; (ii) impact assessments; (iii) post-implementation review; and (iv) design of measures.

[2] TESSD stakeholders include: Basel, Rotterdam and Stockholm Conventions (BRS); Convention on International Trade in Endangered Species of Fauna and Flora (CITES); CUTS International; Ellen MacArthur Foundation; Food and Agriculture Organization (FAO); International Chamber of Commerce (ICC); International Institute for Sustainable Development (IISD); International Monetary Fund (IMF); International Organization for Standardization (ISO); International Trade Centre (ITC); Organisation for Economic Co-operation and Development (OECD); Quaker United Nations Office (QUNO); Forum on Trade, Environment, & the SDGs (TESS); United Nations Environment Programme (UNEP); United Nations Trade and Development (UNCTAD); United Nations Economic Commission for Europe (UNECE); United Nations Industrial Development Organization (UNIDO); World Customs Organization (WCO); World Economic Forum (WEF); World Intellectual Property Organization (WIPO); and World Bank.
[3] See https://www.wto.org/english/tratop_e/tessd_e/tessd_e.htm.

- Analytical Summary of Discussions on Environmental Goods and Services and Renewable Energy: Identifies indicative lists of renewable energy goods (e.g. photovoltaic cells for solar energy, gearboxes for wind turbines, generators for hydropower, and electrolysers for green hydrogen production) and services (e.g. engineering, testing and analysis, environmental consulting, operation, maintenance and repair, and recycling). The document outlines key trade barriers and supply chain bottlenecks, highlights developing country perspectives and opportunities, and examines possible approaches to promote and facilitate trade in these goods and services.

- Mapping Exercise: Trade and Trade Policy Aspects along the Lifecycle of Products: Maps trade and policy relevant aspects across circular economy initiatives and identifies possible trade-related actions in areas such as transparency; standards and regulations; trade facilitation; waste management; capacity building and technical assistance; and technology cooperation.

- Compilation of Experiences and Considerations regarding Subsidy Design: Describes Members' experiences and considerations in the design of subsidies to support the low-carbon transition, including how positive environmental effects can be balanced against potential trade-distorting impacts. It also examines how subsidy design choices may disproportionately affect developing countries and least developed countries (LDCs), including through market distortions.

1.9. A key focus of TESSD has been to identify priorities from a development perspective. At its high-level plenary meeting in December 2024, Members discussed how developing countries are integrating trade and environmental sustainability into national policy frameworks, underscoring the importance of knowledge sharing, the adoption of practical solutions, and capacity building to address gaps in resources and expertise. Participants also examined how multilateral trade processes can create opportunities for developing economies to contribute to global climate and environmental objectives, reinforcing a shared understanding of the role of trade in addressing environmental challenges.

TESSD as an incubator

With 79 Members now co-sponsoring TESSD, the initiative – alongside other WTO trade and environmental sustainability initiatives such as the Dialogue on Plastic Pollution and Environmentally Sustainable Plastics Trade and Fossil Fuel Subsidy Reform Initiative – has played a key role in advancing discussions on the mutual supportiveness of trade and environment. TESSD has helped catalyse engagement for the revitalization of the CTE and strengthened related work across other WTO bodies.

In particular, TESSD has served as an incubator for ideas on trade and environmental sustainability, generating analytical work and practical insights that has informed discussions in a range of WTO bodies. Key contributions include work on the interoperability and transparency of TrCMs, trade and climate challenges and opportunities for developing countries, regulatory cooperation , classification of environmental services, and sustainable agriculture practices and related policies.

"*TESSD is a forum to share experiences, learn and borrow from replicable solutions and build domestic capacity through knowledge exchange*" – Ambassador Matthew Wilson of Barbados

"TESSD has been a key initiative to address environmental sustainability within the multilateral trade framework, fostering the exchange of experiences and challenges that capture the diverse realities of its participants, generating high-value discussions that have enriched the work developed at WTO." – Ambassador Sofia Boza of Chile

"TESSD is a trailblazer at the WTO. You are searching for practical solutions and concrete actions to catalyse the trade and environment agenda. You are breaking down silos and cooperating across traditional structures and fields of expertise to find solutions to global problems" – WTO Director-General Ngozi Okonjo-Iweala

1.10. At MC14, the work of TESSD co-sponsors will be showcased through the technical outcomes developed by the different WGs. The Annex to this publication presents key elements from these outcomes, along with links to the documents in full.

1.11. The following sections provide a crosscutting overview of the substantive knowledge generated during TESSD's first five years. While not exhaustive, this overview illustrates how focused and collaborative engagement can generate valuable insights for policymakers, experts, and the broader public on the role of trade and trade policy in supporting sustainable development and advancing environmental sustainability goals.

2 TRADE IN SUPPORT OF CLIMATE AND ENVIRONMENT: INSIGHTS FROM TESSD[4]

2.1. TESSD work has covered a wide range of topics, from the role of environmental goods and services in advancing the clean energy transition and supporting climate adaptation in sectors such as agriculture and water, to national experiences and international cooperation on carbon measurement standards and other trade-related policies for reducing industry emissions.

2.2. Members have also examined subsidy reform issues and how subsidies may be designed to benefit the environment while minimizing trade distortions, as well as the role of trade and trade policy in fostering a resource-efficient circular economy. This section is organized around three key climate topics that have received considerable attention in TESSD: the clean energy transition; reducing emissions of industry and transport; and climate adaptation, water management, and sustainable agriculture.

2.3. The section draws on Members' discussions and experience sharing, stakeholder contributions, and TESSD outcome documents. Box 1 below provides a few crosscutting, non-exhaustive insights drawn from TESSD's extensive discussions and work. These reflect overarching initial observations that illustrate the value of continued dialogue on trade and sustainability issues.

Box 1 – Crosscutting insights from TESSD work

(1) Dedicated structured discussions on trade and environmental sustainability strengthen knowledge sharing and stakeholder engagement

- **TESSD has enhanced knowledge sharing and dialogue, generating insights relevant for domestic policymakers and stakeholders**. Its work has supported and complemented multilateral efforts, contributing to the revitalization of discussions on trade and climate change in the CTE, informing the work of other WTO bodies, and supporting actions by Members - individually, through enhanced cooperation, or collectively.

- **Engagement from a broad range of stakeholders –** including academia, intergovernmental organizations, private sector, non-governmental organizations, and civil society – **is key to ensure technical accuracy and constructive exchange of views**. Such engagement supports informed policymaking, helps identify gaps, and highlights opportunities for further action and cooperation.

(2) Trade can support environmentally sustainable development when underpinned by coherent and mutually supportive policies across jurisdictions

- **Trade can support the achievement of environmental sustainability.** TESSD discussions have demonstrated that trade already contributes to Members' environmental sustainability objectives and that significant potential exists to further expand sustainable trade. This includes identifying sustainable comparative advantage including for SMEs and developing Members, and pursuing concerted efforts to facilitate trade and create economies of scale.

- **To further harness trade's role in enabling environmental sustainability, policies need to be coherent, interoperable, and mutually supportive.** Experiences sharing has helped build mutual understanding, enabled peer learning, and identified opportunities to enhance coherence and interoperability in key and emerging policy areas, including at the regional level. Discussions were held on aligned standards and shared understanding of key

[4] This section provides an illustrative and non-exhaustive summary of the work undertaken by TESSD. It is a non-binding document, and the considerations and Member practices included in this document are without prejudice to Members' views, rights and obligations under the WTO.

concepts and benchmarks relevant to sustainable trade such as circular economy and other sustainable consumption and production approaches, EGS for lower emissions and renewable energy, production processes, low-carbon transport technologies, and sustainable agriculture.

(3) Transparency remains central to strengthening the relationship between trade and environmental sustainability

- **Improved tracking of key trade flows –** such as critical minerals and renewable energy components, used/refurbished/remanufactured/recyclable goods and scraps – **and greater transparency on product content across the lifecycle** can better inform public policies, particularly in support of EGS facilitation and circular economy initiatives. Achieving this may require reforms, new measures, and enhanced cooperation, including in areas such as the Harmonized System, product passports, export support, or methodologies for calculating water and carbon footprints) while remaining consistent with WTO rules and principles.

- **Greater transparency regarding trade-related measures** adopted for environmental objectives can promote mutual understanding, facilitate peer learning, and support the development of more effective and well-targeted policies.

(4) Addressing development challenges while seizing new opportunities is critical for the low-carbon transition

- **Developing countries and SMEs face a range of trade-related challenges.** Many of the technological solutions identified are capital-intensive and require significant upfront investments, underscoring the importance of facilitating access to finance. Additional constraints include insufficient transport and quality infrastructure (e.g. metrology, certification facilities), small market sizes, low value addition, and complex and costly regulatory requirements add further constraints. Targeted technical assistance, capacity building, enhanced market access, and technology transfer and dissemination were mentioned as essential to support low-carbon transitions.

- **At the same time, the low-carbon transition offers significant opportunities for developing countries and LDCs, which trade can help unlock.** The costs of many relevant technologies – particularly renewable energy – have declined substantially. Opportunities therefore exist for developing countries and LDCs to integrate into low-carbon global value chains at higher value-added stages, develop sustainable comparative advantages (e.g. solar and wind energy generation potential), and reduce energy costs. Expanding sectors such as sustainable transportation fuels, low-carbon industrial solutions, and sustainable agriculture also offer key opportunities.

2.1 Clean energy transition

2.4. TESSD's work has highlighted numerous examples of how trade can support energy transitions. Discussions focused primarily on solar, wind and hydropower, as the most common renewable energy sources for electricity generation worldwide.[5] Their contribution to total global energy generation is expected to rise significantly in the coming decades, in particular for solar and wind[6], whose recent growth has been strongly supported by trade.[7] Across the discussions, there was a recognition that EGS are essential for the development, installation, operation, maintenance, and decommissioning of renewable energy projects.

2.5. Solar, wind and hydropower generation equipment are often part of complex and long value chains, involving diverse technologies, many of which were examined in TESSD (e.g. photovoltaic cells for solar energy, gearboxes for wind turbines, generators for hydropower, and electrolysers for green hydrogen production). Participants have also discussed parts, components and technologies related to other renewable energy sources, including geothermal and marine energy. Relevant

[5] IRENA (2025), Renewable capacity statistics 2025.
[6] IEA (2025), Global Energy Review 2025.
[7] WTO-IRENA (2021), Trading into a bright energy future.

services include engineering services, testing and analysis, environmental consulting, operation, maintenance and repair, and recycling. Indicative lists of key renewable energy goods and services identified by Members are available in outcome document developed by the EGS WG (see Annex III).

2.6. Members identified a series of barriers to the dissemination of renewable energy technologies and services including: tariffs; challenges in certifying products from foreign manufacturers; local content requirements; diverging supply chain traceability requirements and related standards; vulnerabilities to disruptions in supply chains due to concentration of production as well as dependence on certain minerals and raw material inputs; limited technical capacity and regulatory challenges to integrate solar, wind and hydro energy into electrical grids; lack of skilled labour; financing challenges, especially for SMEs; inadequate transport infrastructure and logistics; lengthy authorization and administrative procedures to build projects; and inadequate definitions of goods supporting environmental goals in trade classification (Harmonized System – HS).

2.7. For example, it was noted that 40% of applied MFN tariff rates of TESSD members across all stages of the offshore wind value chain exceed 5%.[8] Some of the highest rates are generally associated with health and safety equipment rather than goods directly used in the construction of wind turbines. Divergent local certification or testing requirements for solar technologies can lead to duplicative requirements. Limited transport infrastructure was also raised as a bottleneck that slows the movement of components, raises costs and restricts access to key markets.

2.8. At the same time, several opportunities and approaches exist to promote and facilitate trade in these goods and services. These include: applying good regulatory practices, in the adoption of measures, in licensing and authorization procedures for renewable energy-related services, and strengthening cooperation on standards, technical regulations and conformity assessment for renewable energy goods. Enhancing the resilience of global supply chains for renewable energy products, as well as improving market access and trade facilitation could lower costs and encourage the flow of goods, services, technologies, and investments across borders, helping to accelerate the energy transition and achieve emission reduction goals. Addressing challenges faced by developing and LDC Members, including through technical assistance and capacity building, can further support their participation in renewable energy sectors.

2.9. As renewable services are often complementary to renewable energy goods and technologies, poor market access conditions for trade in services can hinder trade in renewable goods and technologies and, more broadly, the development of renewable energy projects. In this context, Members suggested further exploration of up- and downstream services needed across the renewable energy value chain, as well as the relevant modes of supply.

2.10. These issues were also examined extensively in the WG dedicated to TrCMs, where Members discussed how such measures can support the energy transition, including through energy efficiency regulations for consumer and industrial products, carbon pricing schemes covering the energy sector, and support measures for the low-carbon energy development, among others. They also compiled practices for designing TrCMs, related to: (i) transparency and consultations; (ii) impact assessments; (iii) post-implementation review; and (iv) considerations for TrCM policymaking (see Annex I).

2.11. Similarly, the WG on Circular Economy – Circularity examined the importance of efforts to ensure that the rapid expansion of renewable energy technologies is resource-efficient. Beyond facilitating the exchange of renewable energy products and components, trade policy can support circularity by providing incentives for related investment, eco-design, addressing waste management complexities, and developing expertise in recycling and circular manufacturing. Environmental impacts can be mitigated by designing products for long service lifespan, ensuring availability of components, services and skilled labour for the maintenance and repair, enabling proper disposal, and reuse where feasible, of renewable energy components.

2.12. TESSD participants also discussed trade-related circularity efforts in the batteries and electronics sector and their relevance for energy transitions. Lithium-ion batteries and other battery types such as solid-state batteries and sodium-ion batteries play an important role in the energy transition, particularly in the context of storing electricity produced from renewable energy and the

[8] United Kingdom (2024), "Offshore Wind Energy: Technical paper by the United Kingdom" (INF/TE/SSD/W/26/Rev.1).

electrification of transport. However, such batteries are resource-, energy- and emissions-intensive to produce. While repurposing and refurbishing through second-life applications of electric vehicle (EV) batteries for energy storage offer promising solutions, challenges still remain. Further relevant trade aspects and practices discussed in the WG on Circular Economy – Circularity can be found in the outcome document produced by the group (see Annex IV).

Box 2 – Case study on batteries recycling, circularity, trade and environment

Recycling of lithium-ion batteries requires advanced technology and scale to be viable, while lead-acid batteries can be recycled cost-effectively at a lower scale with simpler technology. Lead-acid batteries have very high levels of local recycling in developed economies, but trade exists from countries with insufficient domestic recycling capacity to countries which have developed battery waste management capacity. Small-scale recycling in developing countries can cause environmental and health problems linked to harmful effects of lead. Trade related to battery recycling also intersects with measures taken to protect health and environment, which can include restrictions on movement of waste batteries across borders, as under the Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and Their Disposal.

Several other challenges for battery production and recycling to become more circular were identified, including: regulatory fragmentation, characterized by differences in national regulations, product classifications, and labelling; limited collection, sorting, and recycling infrastructures and technologies for batteries, particularly in developing countries; high costs of implementing circular economy measures, including extended producer responsibility (EPR) schemes; and lack of transparency, making it difficult to distinguish recycled content from virgin materials.

2.13. The dedicated WG on Subsidies discussed, through Members' presentations and experience sharing, a range of subsidy reform-related issues including subsidy design, and how this policy tool is being used with the aim of supporting efforts to reduce emissions of the energy sector, while seeking to identify trade-related effects. Subsidies can incentivize the energy transition and remedy market failures by supporting investment in innovative technologies and their uptake. Domestic experiences shared in the discussions demonstrate that Members have used a range of policy approaches, from investment aids, feed-in premiums, and auctions to regional hydrogen hubs, guarantees of origin schemes, and fiscal incentives under comprehensive frameworks. These measures aim to scale up low-carbon energy deployment, enhance energy efficiency, and reduce reliance on fossil fuels by fostering innovation, sector coupling, and storage capacity. It was also raised that energy transition subsidies which are targeted, proportionate, and transparent can help minimize their potential to distort international trade. Similarly, strengthened transparency and information sharing can help improve collective understanding and coherence in the use of subsidies for low-carbon energy development.

2.14. At the same time, TESSD participants noted the importance of minimizing trade-distorting impacts in relation to subsidies and balancing between positive effects for the transition to a low-carbon economy and the distorting effects on trade. The importance of subsidy reform designed to eliminate or reduce adverse environmental impacts of subsidies was also noted. They also discussed that clearly defined market failures, avoiding the provision of undue advantages to specific firms or sectors, adopting technology-neutral criteria, offering open eligibility and competitive allocation processes, and using mechanisms ensuring that subsidies remain temporary, together with rigorous impact assessment and market-based tools, are all practices that can help preserve competitive markets while achieving environmental goals. Other elements raised included: linking subsidies to specific and measurable outcomes; considering other environmental and socioeconomic impacts; and considering potential disproportionate effects of market distortions on developing and LDCs. Further examples of elements for the design of subsidies related to the low-carbon transition are included in the Subsidies WG outcome document (see Annex II).

2.15. Finally, TESSD work has highlighted that the low-carbon energy transition also provides an opportunity for developing countries to take advantage of their sustainable comparative advantage, using abundant renewable energy such as solar, wind, or sometimes geothermal energy, and generating new and low-carbon industrial activity. It is also an opportunity for LDCs to close the energy access gap. Increasing investment is important to the development of the renewable energy sector, particularly in developing countries. For developing countries, establishing and implementing their quality infrastructure systems, which includes standardization, accreditation, conformity assessment, metrology and market surveillance, is essential for enhancing their participation in international trade in renewable energy goods and services.

Box 3: Illustrative examples of Members' experiences and practices presented at TESSD related to the clean energy transition[9]

Chile: Carbon tax

Since 2017, Chile imposes levies on fixed sources from the energy and industry sectors that emit large amounts of carbon dioxide. The implementation of the carbon tax required the deployment of measurement, reporting, and verification (MRV) systems on large facilities. Since 2023, the tax can be offset by presenting eligible emission reduction certificates.

Japan: Energy efficiency efforts

Pursuant to the 'Top Runner Program', Japan requests producers, processors, and importers to deliver energy-consuming equipment and building materials with a certain energy-saving performance based on the most efficient product currently available. Baseline performances are revised periodically.

Switzerland: Energy Strategy 2050 – Promotion of energy efficiency and renewable energy

Promotion systems to increase the use of renewable energy amount to CHF 1.3 billion per year, financed through surcharges on network costs, and cover measures such as one-time investment aids (between 20% to 60%), feed-in premiums and auctions for large photovoltaic (PV) plants. These support measures are traditionally provided on a non-discriminatory basis and are to be phased out by around 2030-2035. They are currently available for eligible technologies such as solar PV, including rooftop and alpine installations; biogas, wood and waste incineration plants; and plants for wind, hydro and geothermal energy.

Examples of trade agreements and cooperative initiatives that include EGS related to climate change mitigation and/or adaptation

- Agreement on Climate Change, Trade and Sustainability (ACCTS) between Costa Rica, Iceland, New Zealand and Switzerland.
- Asia-Pacific Economic Cooperation (APEC) Reference List of Environmental and Environmentally Related Services.
- EU-New Zealand Free Trade Agreement.
- Singapore-Australia Green Economy Agreement.

Battery transparency efforts

China has implemented a battery ID system with a standard that enables traceability.

The EU's digital product passport (DPP) enables supply chain actors to share product data, supporting circular business models. Battery passports can break down information silos, improve recycling and reuse, and increase transparency on carbon impacts.

2.2 Decarbonizing industry and transport

2.2.1 Energy-intensive industries (e.g. steel, aluminium, cement, construction and buildings, fertilizers)

2.16. TESSD discussions have explored trade-related aspects of efforts to decarbonize energy-intensive industries in a variety of ways. For example, discussions in the EGS WG covered relevant technologies being explored to decarbonize industrial processes that involve chemical reactions and

[9] An extensive list of Members' experiences and practices presented at TESSD meetings can be found in the MC14 outcome document of each WG.

high-temperature heat for which the direct use of electricity is a challenge such as the use of green hydrogen. Renewable energy can be converted to green hydrogen via electrolysis. Green hydrogen can thus act as a link between renewable electricity generation and such hard-to-abate sectors, offering a solution to decarbonize some applications, including those where fossil-fuel-based hydrogen is used today.[10] Other EGS relevant to the decarbonization of energy-intensive sectors were also explored, in particular in the buildings and construction sector, where goods and services relevant for the energy efficiency of buildings were identified, such as cooling and heating systems as well as engineering and construction.

2.17. A series of challenges and opportunities to access technologies related to efforts to reduce emissions of energy-intensive industries and sectors exist, in particular for developing countries and LDCs. These include: capital-intensive and high upfront financial costs; limited access to finance and investment opportunities; weak or inadequate infrastructure (e.g. faltering or insufficient electrical grids); limited access to technology and shortage of specialist know-how and skilled workers; complex or unclear regulatory environments; and limited institutional capacity and lack of robust quality infrastructure – which includes facilities to develop, test and certify standards, technical regulations, and conformity assessment procedures.

2.18. These challenges could be addressed by enhanced access to international finance to procure environmental goods and technologies, strengthening international cooperation through technical assistance and capacity building to help develop a skilled local workforce and invest in robust quality infrastructure, as well as including developing countries in the development of international standards to help incorporate their perspectives into the global framework. Additionally, regional integration can help create larger markets, providing the necessary scale for new sustainable industries to become competitive, and strengthen the position of developing Members in global trade.

2.19. In the TrCMs WG, measures such as carbon measurement standards and measures to reduce emissions were identified as a part of the efforts to support the reduction of energy-intensive industries' emissions. Members shared numerous examples of TrCMs covering the sectors of iron and steel, fertilizers, aluminium, hydrogen, cement, as well as construction and buildings. Members considered the implications of the proliferation of carbon measurement standards and the challenges faced by developing Members. Discussions underscored the importance of cooperation in developing common approaches for calculating embedded emissions and in enhancing interoperability of carbon measurement standards. Such discussions continued to develop in other WTO bodies, including in the CTE, where specific thematic sessions were held and Members' submissions tabled proposing ways forward.

2.20. Subsidies and other public support were similarly discussed in the dedicated WG as instruments that can play a constructive role in enabling energy-intensive industries such as steel, aluminium, cement, and chemicals to transition to low-carbon production. Experience sharing revealed diverse policy instruments, including carbon pricing, green bonds, tax incentives, and research and development (R&D) financing, aimed at accelerating industrial emissions reduction and low-carbon technology investment.

2.21. It was also noted by some Members during the experience sharing that transparency and adherence to WTO rules is critical to ensuring that industrial subsidies achieve environmental goals while minimizing trade distortions. Evidence from analytical work by international organizations explored in TESSD meetings indicated that reforming inefficient or poorly targeted subsidies can deliver both climate and competitiveness benefits. Effective subsidy frameworks integrate climate mitigation with economic, social, and development goals, ensure fairness and non-discrimination, and provide access to financial and technical assistance, particularly for developing countries seeking to scale up industrial transformation.

2.22. On buildings and construction, strategies shared in the WG on Circular Economy – Circularity to reduce emissions of the sector included reducing material use, increasing circularity, promoting bio-based and/or regenerative materials, and assessing and reducing emissions from the use of conventional materials like cement and steel. Trade and trade policies can facilitate access to circular technologies and materials, harmonize standards, and promote transparency through tools like

[10] WTO-IRENA (2023), International trade and green hydrogen: Supporting the global transition to a low-carbon economy.

materials passports and sustainability certifications, while tracking their global flows. Trade measures promoting circularity in construction included standards, certifications and sustainable public procurement.

Box 4: Illustrative examples of Members' experiences and practices presented at TESSD related to energy-intensive industries[11]

Republic of Korea: Carbon footprint system

Korea has adopted a joint public and private carbon footprint system for the product lifecycle emissions MRV process on steel and other products.

European Union: Emissions Trading System (ETS)

The EU Emissions Trading System (ETS), launched in 2005, requires polluters to pay for their greenhouse gas (GHG) emissions. It covers emissions from, *inter alia*, the electricity and heat generation, industrial manufacturing and aviation sectors – which account for roughly 40% of total GHG emissions in the EU. It operates in all EU countries plus Iceland, Liechtenstein and Norway, and is linked to the Swiss ETS (since 2020). The sectors covered include: electricity and heat generation; energy-intensive industries (oil refineries, steel works, and production of iron, aluminium, metals, cement, lime, glass, ceramics, pulp, paper, cardboard, acids and bulk organic chemicals); aviation (within the European Economic Area and departing flights to Switzerland and the UK); and maritime transport (50% of emissions from voyages starting or ending outside of the EU and 100% of emissions from voyages between two EU ports and when ships are within EU ports).

Colombia: Green Aluminium Transformation Complex (GALTCO) project

Colombia's Green Aluminium Transformation Complex (GALTCO) project is an initiative to produce near-zero carbon aluminium, illustrating the role of targeted government policies in advancing green industry. Colombia's green energy system – with abundant hydroelectric power and renewable sources – forms an ideal foundation for lower emission aluminium production. The initiative represents a USD 1.1 billion investment to establish a low-carbon aluminium production facility with an installed capacity of approximately 350,000 tons per year. By focusing on climate action, job creation, and regional development, GALTCO seeks to address a major aluminium supply deficit in the American market while promoting Sustainable Development Goals. With production expected to begin in 2029 and reach full capacity by 2030, GALTCO, in partnership with EDF (Electricité de France), aligns with Colombia's National Development Plan 2022-2026, promoting sustainable economic growth and transforming linear economic models into circular systems.

2.2.2 Transport

2.23. Given the intrinsic connection between international merchandise trade and the transport sector, TESSD discussions covering trade-related efforts to reduce emissions from international freight can provide relevant insights for trade experts. For example, a series of alternative fuels and related technologies (e.g. for their production, conversion or transportation) were raised in the EGS WG as potentially supporting reduced emissions efforts, including (green) hydrogen and ammonia, sustainable aviation fuels (SAF), bioethanol, renewable and biodiesel, and biomethane.

2.24. On biofuels, discussions noted how they could be produced from a wide variety of feedstocks and different processing technologies. Biofuels can, *inter alia*, be derived from dedicated crops or wastes and agricultural residues, with additional considerations related to the biofuel's sustainability and GHG savings, as well as the balance between biofuel production and environmental protection.

2.25. Discussions explored how the development of the biofuels sector using sustainable feedstocks suitable to each country can contribute to energy security, economic growth and bring value addition to farmers while also resulting in valuable co-products and decreased dependence on fossil fuels.

[11] An extensive list of Members' experiences and practices presented at the TESSD meetings can be found in the MC14 outcome document of each WG.

There are opportunities presented by biofuels for developing countries to be integrated into global value chains as exporters of renewable energy inputs, and to benefit from trade. Developing countries similarly have opportunities to leverage comparative advantages in natural resource abundance to attract investment and develop green hydrogen production thereby contributing to economic diversification, reducing dependence on traditional industries. Regional trade agreements like the African Continental Free Trade Area (AfCFTA) could create economies of scale to foster participation of developing and LDC Members in global value chains as well as regional value chains.

2.26. At the same time, inadequate transport infrastructure and logistics in themselves were raised as barriers to further deployment of EGS. Limited transport infrastructure can create bottlenecks by slowing the movement of components, raising costs and limiting access to key markets. Wind and hydropower goods are often large and heavy. Transportation challenges, including long-distance shipping, limited infrastructure, and regulatory constraints, can increase costs and cause delays. Port infrastructure is similarly crucial for the offshore wind industry, both during installation, operation and maintenance. Delays to port upgrades can cause shortages of wharf space and incur additional charges.

2.27. TrCMs also play an important role in the development of the green hydrogen sector, which could support the reduced emissions of the transport sector in the future. For example, efforts to further align standards, conformity assessment procedures and regulatory frameworks along the hydrogen value chain can support this sector. This can be complemented by reducing operational complexities and improving business environments to support investment, promoting renewable energy sources that are essential to produce renewable hydrogen, and interoperable certification schemes.

2.28. In the dedicated WG, subsidy reform and targeted support were discussed as having the potential to accelerate the reduction of emissions of global transport systems, particularly in maritime and aviation sectors. In maritime transport, a combination of regulatory and financial mechanisms – such as national and regional programmes supporting sustainable marine fuels, vessel retrofitting, and sustainable ports – illustrate how coherent policies can drive the transition toward zero-emission shipping. In aviation, sustainable aviation fuels (SAF) represent a pivotal pathway for emission reduction. Policy frameworks demonstrate how targeted incentives, blending mandates, and R&D financing can scale up SAF production and narrow its price differential with fossil fuels. Harmonized standards, lifecycle assessment, and cooperation with developing countries are key to ensuring environmental integrity and equitable participation in low-carbon transport transitions.

2.29. Circularity aspects of efforts to electrify transport systems were also explored in the WG on Circular Economy – Circularity. Both upstream (e.g. product design and labelling, tracking materials across their lifecycle) and downstream/end-of-life policies (end-of-use and end-of-life battery management) are important for circularity to support the electrification of transport. Domestic programmes shared included renewable fuels obligations, circular economy policies such as battery passports, and EPR systems. Downstream, experiences included recycling batteries and waste management regulations. The challenges of small economies in achieving scale highlighted the need for regional cooperation, and trade tools – such as centralized/regional collection and processing facilities, Green List systems to fast-track responsible recyclers, and relying on existing regional economic commissions – as potential platforms for cooperation.

Box 5: Illustrative examples of Members' experiences and practices presented at TESSD relating to transport[12]

Canada: Support for the development of low-carbon hydrogen

The Canadian Government launched the CAD 1.5 billion Low-carbon and Zero Emissions Fuels Fund in December 2020 to support the production of low-carbon fuels, including hydrogen, and reduce emissions in the transport, marine, and aviation sectors. Additionally, funding for research and development of low-carbon hydrogen production technologies and infrastructure has been provided by the government.

Switzerland: Tax relief on biofuels

To reduce CO2 emissions in the transport sector, tax reliefs are granted to importers and exporters according to compliance with sustainability requirements. The tax relief was granted to importers and manufacturers who could demonstrate that their biofuel met ecological and social requirements, including that emissions were at least 40% lower than those from fossil fuels; environmental impacts were not significantly higher than those of fossil fuels; production requirements that did not require the conversion of carbon-rich land; and that production only occurred on legally acquired land through socially acceptable production conditions in accordance with the fundamental conventions of the International Labour Organization (ILO).

Norway: Incentives provided for purchase and usage of electric vehicles (EVs)

Norway has supported the introduction of EVs for some time. Policies do not discriminate between countries of origin or producers but between cars with combustion engines and EVs. Incentives included the exemption of certain taxes at the moment of purchase of new EVs, most notably the special 'one-off registration tax' on vehicles and the 25% value-added tax (VAT). Support measures for usage have also been granted, such as exempted or lower fees for road toll and ferries, subsidized public charging facilities, as well as privileged access to road lanes generally reserved for buses and taxis. These incentives are considered to have contributed substantially to the steady increase in the number of EVs sold in the Norwegian market, as a proportion of total car sales, over the last decade. Over time, Norway has reduced tax incentives for the purchase and usage of EVs, reflecting the increasing technological maturity and competitiveness.

Brazil: National Aviation Biofuel Program (ProBioQAV)

The legislative proposal for a "Fuel of the Future" bill foresees the establishment of a National Aviation Biofuel Program (ProBioQAV). The Program would aim to carry out improvements in logistics, which are critical for the sustainability of fuel distribution and for reaching the ultimate goal of reduced emissions. According to the proposed law, airlines will be required to reduce GHG emissions by 1% annually starting in 2027 by gradually increasing the percentage of SAF added to kerosene, until reaching a total drop of 10% in 2037.

Efforts to promote EV batteries circularity

Rwanda adopted and incentivized electric mobility in April 2021, including actions such as developing fiscal incentives for EVs and standards for imported vehicles. These efforts aim at minimizing pollution and waste in transport, and are complemented by actions to extend product lifecycles and promote the repurposing of physical and natural assets.

As cells and batteries imports rose since 2020, especially from hybrid or EV imports, Mauritius included in its Circular Economy roadmap and Action Plan (2023-2030) a priority on batteries recycling, highlighting challenges for small economies in achieving scale in recycling lithium-ion batteries.

[12] An extensive list of Members' experiences and practices presented at the TESSD meetings can be found in the MC14 outcome document of each WG.

2.3 Climate adaptation, water management and sustainable agriculture

2.30. Climate adaptation, in particular the challenges imposed by global warming on agriculture productivity and water shortages, were also explored in TESSD meetings. Integrated water resource management is critical to ensure a sustainable supply of water, water quality, and mitigation of water-related disasters in the face of changing climate when the frequency and severity of extreme weather events increase. When discussing key goods and technologies in the EGS WG, Members, *inter alia*, mentioned drippers and sprinklers for irrigation solutions, water meters, filtration systems, leak detection technology, water treatment and desalination technologies, rainwater harvesters and fog traps, data management and digital technologies (e.g. sensors, data analytics, smart water solutions), and quality water conveyance systems (e.g. pipes, valves, pumps and storage).

2.31. Technologies for climate change adaptation in agriculture cover a range of solutions, including climate-resilient farming measures, crop management strategies, agricultural water management, and environmental monitoring. Members mentioned several technologies, goods and services that could support sustainable agriculture and climate adaptation, including some fertilizers (slow-release, bio-bacterial, organic), precision agriculture tools (sensors, drones, and smart irrigation systems), solar-powered cold storage infrastructure, biotechnology (climate-resilient crop varieties and gene editing techniques), biodiversity-based production models (e.g. agroforestry systems), farm advisory services, engineering, and R&D. It was noted that the identification of some of these goods and practices as supporting sustainable agriculture depends on their use, context-specific production and use conditions. For instance, plant growth regulators (e.g. herbicides, anti-sprouting products) can have a damaging impact on the environment. Similarly, the positive impact on the environment of drones largely depends on the use, chemicals sprayed, and alternatives; the same goes for fertilizers, or drippers and sprinklers.

2.32. Additionally, effective climate services play a key role in enabling better planning for resilient agricultural practices through improved forecasting and the identification of suitable crop varieties. These services also support the design of robust infrastructure and contribute to sustainable agricultural financing. Moreover, climate services are essential for adapting and supporting mitigation responses, enhancing food security, informing trade and subsidies, guiding research on resilient varieties, and integrating into agricultural finance and insurance while emphasizing stakeholder engagement and effective communication.

2.33. Technology needs assessment projects in developing countries identified water and agriculture as two priority sectors for climate change adaptation. Priority technologies for water included irrigation systems (e.g. drip irrigation), water storage (e.g. bladder tanks), water harvesting (e.g. rooftop rainwater) and water management (e.g. smart metering, leakage detection and repair, pressure management). Challenges to the uptake and diffusion of water management technologies include: the absence of local suppliers; lack of know-how and skills for installation, repair and maintenance; limited absorption capacity; lack of standards and certification; costs (including transport and tariffs); and complexities regarding software and data ownership.

2.34. Some Members highlighted that tackling climate adaptation in sustainable agriculture requires a coherent package of policies, including trade in environmental goods and services as an important element, helping farmers cope with the challenges of climate change, enhance agricultural productivity and reduce resource utilization and negative environmental impacts. Developing countries face the need for investment in infrastructure, such as solar-powered cold storage and improved logistics, as well as capacity building and technical assistance for the adoption of technologies in line with their development needs. Some additional challenges for developing countries raised in discussions include the creation of non-tariff barriers due to differing standards, the variability of emissions in agriculture due to differences in production techniques, climate and national circumstances, and the need for enhanced capacity building and market access.

2.35. It was noted that reducing emissions from the agriculture sector can also include efforts to utilize TrCMs to reduce emissions of fertilizer production and use, including promoting low-carbon hydrogen that support the production of green ammonia for nitrogen fertilizers. Some of the measures discussed included, *inter alia*, carbon pricing and border carbon adjustments, product standards, regulations to reduce emissions from transportation, and funding for low-carbon technology research.

2.36. Other opportunities and approaches for trade policy raised in discussions relate to, *inter alia*: cooperation on water efficiency standards and labelling; increase coherence and reduce divergence of sustainability-related measures; enhancing of value chain traceability, sustainability standards, certification and product stewardship , taking into account local conditions and avoiding excessive compliance burdens particularly for developing countries and MSMEs; inclusion of goods and services in trade agreements that foster sustainable agriculture as well as sustainable production and consumption; collaboration on technical assistance, capacity building and technology transfer on voluntary and mutually agreed terms; public-private partnerships and private sector collaboration to facilitate investment, innovative solutions and efficient services delivery; and encouragement of innovation and application of new technologies such as precision agriculture and digital tools for enhanced productivity, and promotion of international cooperation.

2.37. Subsidy reform in agriculture was noted by some Members as a lever for promoting climate adaptation, biodiversity conservation, and sustainable food systems. Analytical work and country experiences discussed indicated that market price support, output, and input subsidies may be environmentally harmful, whereas spatially targeted agri-environmental payments can yield positive outcomes for soil health, water conservation, and ecosystem resilience. Innovative policy design combines measures to mitigate negative environmental effects with incentives for sustainable production, R&D on climate-resilient crops, and direct payments linked to environmental performance. Transparency, monitoring, and science-based impact assessment enhance the effectiveness of such measures while promoting policy coherence across agriculture, trade, and biodiversity frameworks

2.38. Finally, during discussions on circular economy initiatives in the building and construction sectors, some relevant elements were also raised. Efforts on the topic focus on avoiding waste, building with less, as well as a transition towards bio-based building materials, in order to transition to low-carbon buildings. Trade-related aspects in this transition to bio-based building materials include the standardization and certification of bio-based materials, as well as the mainstreaming of alternative materials in conventional construction. Sustainable public procurement was noted as a powerful tool to move the market towards sustainable practices. Trade solutions discussed would support the uptake of nature-based products that were technically suited for construction, adequate when adapting to extreme weather events, and were grown and produced in a sustainable manner.

Box 6: Illustrative examples of Members' experiences and practices presented at TESSD relating to climate adaptation, water management and sustainable agriculture[13]

Innovative mechanism regarding sustainable production of palm oil in the Comprehensive Economic Partnership Agreement (CEPA) between the European Free Trade Association (EFTA) States and Indonesia

A relevant innovation of the CEPA, particularly for sustainability, was the establishment of a link between the preferential tariff rate quotas for Indonesian palm oil granted under the CEPA and the sustainability criteria set out in the chapter on trade and sustainable development. In the absence of a universally accepted standard for the sustainable production of palm oil, parties to this agreement had decided to rely on well-established voluntary sustainability standards (VSS). A certification according to selected VSS is recognized by Switzerland as sufficient proof of respect of the CEPA criteria.

Kingdom of Saudi Arabia: Saudi Green Initiative and Middle East Green Initiative

Under the Saudi Green Initiative (SGI) and Middle East Green Initiative (MGI), the Kingdom of Saudi Arabia aims to deploy low-carbon technologies and innovative approaches in collaboration with other countries, regional partners and multilateral development banks, including in the agriculture, energy and waste sectors. Other programmes aimed at addressing climate change challenges include the Circular Carbon Economy National Program, the National Renewable Energy Program, and the Saudi Energy Efficiency Program.

Costa Rica: NAMA Café

The NAMA Café initiative seeks to promote low-emission and sustainable coffee production and processing in Costa Rica, through the adoption of low-emission technologies and the efficient use of water and energy. An incentive mechanism by the Costa Rican Coffee Institute (ICAFE) allows NAMA Café beneficiaries to receive a monetary contribution for a maximum of three investment projects of up to 10% of the investment. Incentives might be obtained for technologies related to wet and dry milling, by-product treatment system, or use of renewable energy, among others. The development of these programmes entails an exhaustive process of consultation at the domestic level with government sectors and entities, including the Ministry of Foreign Trade, which ensures that any state policy adheres to the rules of the multilateral trade system and international law.

Paraguay: Good agricultural practices

In Paraguay, about 65% of farms carry out some soil management and conservation practices, of which 64% apply crop rotation and 16% direct sowing, which are the most frequent good agricultural practices. Organic production is not automatically considered a good agricultural practice as it involves trade-offs (e.g. mechanical processes that harm soil) that may not lead to a net-positive impact on the environment. Other good agricultural practices, *inter alia*, include the use of biotechnology which enhances productivity and lowers the use of resources, while precision agriculture provides for a more efficient use of equipment and plots, lowering environmental impact. In terms of farming practices, the adjustment of animal load according to the intrinsic capacity of available fields will enhance productivity and can bring the greatest benefits for climate change mitigation.

Environmental impacts of sustainable agricultural policy reform

The UK estimates the Environmental Land Management Schemes deliver high value for money for every £ invested (where high value for money is a benefit-cost ratio over 2 as reported by the UK's National Audit Office), with environmental benefits to habitats and species, water quality and carbon reduction. The Environmental Land Management Schemes include the Sustainable Farming Incentive (SFI), Countryside Stewardship Higher Tier (CSHT) and Landscape Recovery schemes.

Israel considers that there are clear positive environmental effects of agricultural subsidies. For instance, subsidies for integrated pest management drastically reduce chemical pesticide use while maintaining the same level of production; and subsidies for soil conservation lower nitrous oxide emissions, reduce water evaporation and decrease soil erosion, while increasing organic matter, biota and wildlife presence.

3 CONCLUDING REFLECTIONS FROM THE CO-CONVENORS

3.1. Over the past five years, TESSD has addressed several key interlinkages between trade and environmental sustainability. This work is widely recognized as having contributed to building significant technical knowledge on how trade can support environmental sustainability, covering a wide range of topics regarding EGS, TrCMs, Circular Economy – Circularity, and Subsidies.

3.2. Reflecting on TESSD's nature of work at the WTO, its trade expertise, and its "comparative advantage" in bringing together diverse perspectives through structured, transparent and inclusive dialogue, discussions have focused on the specific role trade and trade policies can play in supporting objectives relevant to its co-sponsors. These include, *inter alia*, the energy transition, water management, sustainable agriculture, sectoral value chains, resource-efficient and circular economies and climate adaptation. In this regard, TESSD has proven to be a valued and complementary incubator for innovative ideas, which, once sufficiently mature, can support Member-led discussions in other WTO workstreams and committees, be taken into account by domestic and regional policymakers, or inform other international processes and discussions.

3.3. TESSD has also strengthened dialogue among WTO Members by bringing together trade, environment and other officials to jointly identify relevant challenges, opportunities and existing efforts, and to present and discuss these at TESSD meetings. Moreover, given the technical complexity of the issues addressed, TESSD has substantially increased stakeholder engagement on trade and environmental sustainability. Participants have benefited from the contributions from a broad range of stakeholders, including the private sector, international organizations, academia and NGOs, offering local, domestic, regional or international perspectives.

3.4. Collectively, this work has generated the substantive knowledge reflected in the "MC14 Package", providing a robust technical basis to support efforts to respond to the first action trade ministers agreed to pursue in 2021, namely to "identify concrete actions that participating Members could take individually or collectively to expand opportunities for environmentally sustainable trade in an inclusive and transparent way, consistent with their obligations."

3.5. TESSD co-sponsors can take pride in the collective work undertaken and the added value achieved. When TESSD was established, it was done so with a view to reinvigorate discussions on trade and environmental sustainability, and the initiative has largely met that goal. While recognizing that TESSD was established to respond to a shared objective among co-sponsors, its unique set-up has served as a key platform for innovative exchanges and dialogue on trade and environmental sustainability issues. As challenges at the intersection of trade and environmental sustainability persist, we will continue to encourage Members to engage collectively and transparently in identifying effective and constructive ways to ensure that trade and trade policy form part of the solution in areas of common interest to co-sponsors.

[13] An extensive list of Members' experiences and practices presented at the TESSD meetings can be found in the MC14 outcome document of each WG.

ANNEX I: TRCMS WG OUTCOME DOCUMENT – MAIN ELEMENTS

The TrCMs Working Group outcome document (WT/MIN(26)/22/Add.2) provides a compilation and mapping of policy measures shared by Members aimed at achieving specific climate objectives, and the identification of practical ways to enhance cooperation to achieve these objectives. It is a non-binding document without any prejudice to Members' rights and obligations under the WTO Agreements. It provides examples of Members' trade-related climate policies reflecting their national circumstances. They are illustrative of approaches that have been taken and are not intended to be comprehensive or represent an endorsement by Members.

Below, the titles of the main sections of the document are reproduced, followed by their initial paragraphs. More information and concrete examples of policies shared by Members are available in the document.

1 COMPILATION AND MAPPING OF TRADE-RELATED CLIMATE POLICIES

1.1 Carbon pricing and Emissions Trading Systems (ETS)

Carbon pricing is a market-based instrument that sets a price on carbon dioxide (CO_2) or equivalent GHG emissions. Carbon prices encourage producers to decrease the carbon intensity of the production and transportation processes, and consumers to choose less carbon-intensive goods and services. By allowing flexibility in how emissions are reduced, carbon pricing can also stimulate innovation in low-carbon products and production processes. Explicit carbon pricing takes two main forms: a carbon tax or an ETS. A carbon tax sets a price on carbon emissions, influencing producers and consumers to determine the quantity of emissions. Under an ETS, the regulator sets a cap on total GHG emissions and the market determines the carbon price, with operators buying or selling allowances to match their emissions.

1.2 Border Carbon Adjustments (BCAs)

In general, BCAs seek to apply a carbon price per ton of CO_2 on imported goods, reflecting similar costs applied on domestic goods. In economic terms, BCAs therefore turn carbon pricing into a demand-side policy affecting domestic and foreign producers.

1.3 Taxes and incentives

Taxes and incentives are market-based instruments through which governments aim to support the development and deployment of climate-friendly goods and technologies. Incentive policies related to climate change may focus on e.g., increased use of renewable and cleaner energy sources, development and deployment of energy-efficient and low-carbon content goods and technologies, or development and deployment of carbon sequestration technologies.

1.4 Regulatory measures – standards, certifications and labelling

Regulatory measures can be used to pursue climate objectives, for instance by setting requirements for accounting emissions, energy consumption, or energy efficiency. Regulatory measures include standards (voluntary) and technical regulations (mandatory), which set out product characteristics or requirements for their related processes and production methods. Such requirements may be accompanied by implementation and compliance measures, such as labelling requirements and conformity assessment procedures (e.g. testing, verification, inspection and certification).

1.5 Voluntary actions and cooperative initiatives

Voluntary actions and cooperative initiatives are voluntary efforts by multiple stakeholders (public and non-public) to work towards specific climate goals. Voluntary actions can take the form of voluntary agreements between a government authority and private sector parties, where the parties set targets or commitments in lieu of, ahead of, or in conjunction with regulatory measures.

1.6 Trade preferences and sustainability-linked instruments

Trade preferences and sustainability-linked instruments can be used to facilitate market access for specific climate-friendly and environmental goods or goods that meet certain sustainability criteria.

2 AREAS FOR ENHANCING COOPERATION

The section contains a non-exhaustive number of illustrative elements from Members' statements for enhancing international cooperation on TrCMs. These elements relate to four areas: (i) information sharing and dialogue; (ii) fostering convergence on standards; (iii) conformity assessment and certification schemes; and (iv) technical assistance and capacity building.

2.1 Information sharing and dialogue on TrCMs

Establishing mechanisms for regular information exchange on TrCMs among governments, international organizations, industry stakeholders, and civil society enhances transparency, mutual understanding and the effectiveness of TrCMs. In particular, sharing good practices among WTO Members, as well as national and regional initiatives, might serve as a valuable foundation for collective learning. Information that could be particularly beneficial to share includes methodologies for carbon measurement and reporting; national, regional and international approaches to standard-setting; and experiences with TrCMs under consideration or implementation.

A variety of multilateral forums could provide useful venues for such dialogue, including various WTO bodies and initiatives, such as CTE, TBT Committee and TESSD, as well as other relevant international organizations, forums and initiatives.

2.2 Fostering convergence on carbon measurement standards

Advancing coherence and interoperability of carbon measurement and other related standards across sectors is a priority to facilitate regulatory alignment, reduce trade barriers, and better track emission reduction efforts.

The development of international standards for carbon measurement, including sector-specific instruments, can underpin TrCMs, such as carbon labelling and certification, thereby supporting the decarbonization of emission-intensive industries. Referencing international standards in technical regulations might reduce regulatory fragmentation and helps ensure that technical regulations do not create unnecessary obstacles to trade. Members can help advance interoperability and convergence related to carbon measurement and other related standards in various WTO bodies, including the TBT Committee.

2.3 Conformity assessment and certification schemes

Supporting convergence and mutual recognition across Members of conformity assessment procedures related to carbon measurement might contribute to reducing unnecessary trade barriers, incentivizing sustainable production methods and ensuring consistent implementation of TrCMs. In particular, these efforts would be valuable in carbon intensive industries, such as aluminium, cement, iron and steel, and fertilizers.

Moreover, the development of certification methodologies that are interoperable across jurisdictions is also key to facilitate cross-border trade in current and emerging sustainable solutions, such as green hydrogen.

2.4 Technical assistance and capacity building

Enhancing technical assistance and capacity building will be critical in addressing challenges faced by Members at different development levels in relation to the design, adoption and implementation of TrCMs. Enhancing developing Members' capabilities in relation to TrCMs may facilitate effective policy design and implementation, helping to maximize climate and trade benefits and ensure that TrCMs do not create unnecessary obstacles to trade. Technical assistance could help address constraints related to institutional capacity, human resources, and technology gaps, among others.

Specific areas for technical assistance and capacity building might include: the development of carbon measurement standards; creation and maintenance of emissions inventories; design and implementation of TrCMs, such as carbon border measures and labelling schemes; establishment of conformity assessment and certification schemes; and support in data collection and reporting mechanisms.

ANNEX II: SUBSIDIES WG OUTCOME DOCUMENT – COMPILATION OF DESIGN ELEMENTS

The Subsidies Working Group outcome document (WT/MIN(26)/22/Add.5) provides a compilation of experiences and potential considerations regarding subsidy design and provides an illustration of elements that are considered by some Members in the design of subsidies related to the transition to a low-carbon economy. The descriptions under each example of subsidy design elements are aimed at facilitating understanding of underlying considerations and concepts. They are based on Members' discussions and inputs as well as presentations and relevant analytical work by stakeholders, and do not reflect the views of all, or any subset of, Members.

The indicative considerations and the accompanying language are not intended as exhaustive or prescriptive, and compliance with them is not a justification for measures that violate WTO rules on subsidies. The document is non-binding, and the considerations and Member practices included therein are without prejudice to Members' rights and obligations under the WTO agreements.

Below, the "considerations for subsidy design" section of the outcome document are reproduced. More information and concrete examples shared by Members are available in the document.

Rationale and design elements

1. **Providing subsidies aimed at remedying a market failure and incentivizing the transition to a low-carbon economy**: Subsidies can be an appropriate means of addressing market failures, such as externalities, when supporting investments in emerging practices, innovative technologies and R&D, with a clear *incentive to induce* sustainable behaviour. Such subsidies can effectively signal to businesses and consumers the importance of transitioning to a low-carbon economy, for example, and encourage a shift in behaviour and investment patterns towards new technologies or more expensive technologies that achieve a desired outcome.

2. **Providing subsidies proportionate to the desired outcome**: Scaling subsidies in alignment with their specific goals related to a low-carbon transition can enhance efficiency and effectiveness. For example, to promote cost-effectiveness, the subsidy amount could reflect the minimum level necessary to trigger behavioural change, for example by covering only part of the additional costs associated with achieving specific low-carbon objectives such as GHG emission reductions. Proportionality may promote efficient use of public funds and maximize environmental benefits.[14]

3. **Providing subsidies on a non-discriminatory basis**: Ensuring equal access to low-carbon transition subsidies for all eligible firms, regardless of their country of origin, can promote global competition and help the most efficient and innovative low-carbon technologies scale up. Conditioning subsidies on criteria such as emissions reduction potential or technological impact, amongst other socioeconomic considerations, may help prevent trade distortions and cost inefficiencies. Transparent and open subsidy schemes can also reduce knowledge asymmetries and foster international cooperation.[15]

4. **Avoiding export subsidies and local content requirements that are inconsistent with the Agreement on Subsidies and Countervailing Measures, and giving preference to less distortive types of subsidies, where feasible**: Article 3.1(a) of the SCM Agreement prohibits, in law or in fact, whether solely or as one of several other conditions, subsidies contingent upon export performance. Equally, Article 3.1(b) of the SCM Agreement prohibits subsidies contingent, whether solely or as one of several other conditions, upon the use of domestic over imported goods. In practice, tying financial support to domestic input sourcing requirements can create trade distortions and limit global supply chain efficiencies. Local content requirements can raise production costs, reduce competition, and lead to retaliatory measures from trading partners. Less distortive forms of support, such as market-based incentives (e.g. feed-in tariffs, green procurement, or technology-neutral auctions) can stimulate demand, encourage innovation, lower costs, and facilitate deployment of clean technologies.[16]

[14] WTO, IMF, OECD, World Bank (2022), *Subsidies, Trade, and International Cooperation*.
[15] WTO, IMF, OECD, World Bank (2022), *Subsidies, Trade, and International Cooperation*.
[16] OECD (2024), *Green Industrial Policies for the Net-Zero Transition*; OECD (2023), *Government Support in Industrial Sectors*.

5. **Ensuring time limitations for the subsidy programme**: Time limits on low-carbon transition subsidies facilitate prevention of long-term market distortions and industry dependence on government support, while ensuring they are limited to what is necessary to trigger a change in behaviour and address market failures.[17] Mechanisms ensuring that subsidies remain temporary and effective include expiration dates, sunset clauses, gradual phase-outs, and regular reviews that assess the subsidy's effectiveness, allowing for adjustments or termination once its objectives are met.

Impact considerations (economic, environmental and social)

6. **Minimizing trade-distorting impacts when designing subsidies and balancing between positive effects for the transition to a low-carbon economy and the distorting effects on trade**: Seeking that low-carbon transition subsidies are targeted, proportionate, time-limited and transparent can help minimize their potential to distort international trade. Addressing clearly defined market failures, avoiding the provision of undue advantages to specific firms or sectors, adopting technology-neutral criteria, offering open eligibility and competitive allocation processes, and using mechanisms ensuring that subsidies remain temporary, together with rigorous impact assessment and market-based tools are all practices that can help preserve competitive markets while achieving environmental goals. Minimizing trade distortions supports a level playing field and reduces the risk of disputes or retaliatory measures from trading partners. This can also help promotes subsidies that contribute effectively to global climate efforts without undermining multilateral trade rules.

7. **Linking subsidies to specific and measurable outcomes**: Subsidy programmes that are linked to specific outcomes such as those related to decarbonization strategies (e.g. measurable reductions in GHG emissions, improvements in energy efficiency, or deployment of renewable energy capacity) may help promote environmental credibility. This means linking funding to activities, technologies, or infrastructure with a demonstrable contribution to decarbonization. Establishing transparent criteria and requiring emissions reporting can help monitor impact and promote accountability.

8. **The impacts of the programme beyond GHG emissions reductions such as other environmental and socioeconomic impacts**: Low-carbon transition subsidies can have broader benefits, such as improved air quality, biodiversity protection, job creation, or innovation. However, they may also carry risks, such as resource overuse, ecosystem degradation, or unequal access to benefits. Impact assessment in the design of low-carbon transition subsidies, which takes into account environmental and social outcomes, can help maximize benefits while mitigating harms.[18] For some Members, energy security, economic development, and poverty eradication are important considerations to take into account when assessing impacts.

9. **Taking into account how market distortions might disproportionately affect developing and least developed countries as well as different development levels of countries**: Subsidy-induced market distortions can have unequal impacts across countries, disadvantaging developing and least developed countries with less fiscal space or weaker domestic industries. These countries may struggle to replicate similar support measures and compete with subsidized goods. A fair and inclusive approach to subsidy design can support a just transition and help prevent the widening of development gaps. Transparency in subsidy design and implementation, along with international collaboration, can play an important role in identifying potential market distortions and preventing or mitigating related effects on these countries.

[17] OECD (2024), *Green Industrial Policies for the Net-Zero Transition*; WTO, IMF, OECD, World Bank (2022), *Subsidies, Trade, and International Cooperation*.
[18] OECD (2024), *Green Industrial Policies for the Net-Zero Transition*; WTO, IMF, OECD, World Bank (2022), *Subsidies, Trade, and International Cooperation*.

Implementation and governance considerations

10. **Ensuring consistency and predictability in the provision of subsidies**: A stable and predictable policy environment fosters long-term investment and innovation and supports the effectiveness of low-carbon transition subsidies. Consistent rules and funding commitments help investors plan for infrastructure development, technology deployment, and supply chain adjustments. Clear eligibility criteria, application processes, and timelines can enhance confidence in the implementation of a subsidy programmes. Regular stakeholder engagement and impact assessments can also help maintain policy stability while allowing for necessary adjustments over time.

11. **Ensuring coherence, inclusivity and transparency in the provision of subsidies**: Ensuring coherence of low-carbon transition subsidies with other environmental and trade policies could enhance complementarity and support achievement of policy goals. Transparent governance mechanisms, such as open consultation processes, can enhance trust and allow stakeholders to track progress and effectiveness. Inclusivity is also key, aiming to ensure that small businesses, indigenous communities, and/or developing-country producers can benefit from low-carbon transition subsidy schemes.

12. **Managing implementation risks through appropriate institutional frameworks and capacity building**: Effective implementation of low-carbon transition subsidies can be supported by strong institutional frameworks and administrative capacity. Clear expertise, sufficient resources, and coordination between regulatory bodies can enhance efficiency and help avoid conflicting policies. Capacity-building initiatives, particularly in developing Members, can enhance the ability of institutions to design, monitor, and enforce subsidy programmes effectively. Digital tools and streamlined application processes can also reduce administrative burdens and improve efficiency.

13. **Promoting international coordination to promote effectiveness and efficiency**: International coordination can help avoid negative cross-border impacts of subsidies. Platforms such as the WTO, G20, and climate-focused initiatives support alignment of subsidy policies with global sustainability goals. Joint initiatives, such as co-financing clean technology projects or harmonizing subsidy eligibility criteria, can further enhance policy effectiveness and prevent subsidy competition. So can alignment of programmes with international standards, where applicable. Transparent sharing of best practices and subsidy impact assessments can improve policy design and implementation worldwide.

14. **Promoting complementarity with other decarbonization policies**: The effectiveness of subsidies can be increased if they work alongside other decarbonization policies such as technical regulations, carbon pricing and renewable energy targets. Effective policy coherence can enhance overall environmental and economic outcomes and promote that subsidies are effectively minimized while helping industries transition without duplicating or counteracting existing incentives. For instance, targeted subsidies can ease the adoption of clean technologies in response to carbon pricing signals. A well-designed mix of subsidies and market-based mechanisms can incentivize companies to invest in low-carbon solutions without over-reliance on public funding; this mix may reflect the needs of developing countries for flexibility in pursuing low-carbons. Subsidizing domestic producers to comply with certain regulatory requirements may increase the compliance burden on foreign producers. In the same vein, ensuring that subsidies do not conflict with other environmental objectives supports policy integration. Paying attention to the cumulation of subsidies from different instruments for a project or activity helps prevent overcompensation or exceeding of the maximum allowable support under any individual instrument.

ANNEX III: EGS WG OUTCOME DOCUMENT – KEY INSIGHTS

The EGS Working Group outcome document (WT/MIN(26)/22/Add.3) provides an analytical summary of discussions on environmental goods and services, covering: (i) indicative lists of relevant goods and services; (ii) trade barriers and supply chain bottlenecks; (iii) developing country perspectives; and (iv) opportunities and approaches for promoting and facilitating trade in these goods and services. It is a non-binding document, and the considerations and Member practices included in this document are without prejudice to Members' rights and obligations under the WTO Agreements.

The group also developed a document containing "key insights", reflected in the outcome document and whose content is reproduced below. More information and concrete examples of policies shared by Members are available in the full analytical summary.

Key Insights on Environmental Goods and Services

Environmental goods and services (EGS) can support countries in achieving their environmental and sustainable development objectives. Discussions by Members in TESSD have focused on climate mitigation and adaptation goals, covering priority sectors such as renewable energy, water supply and wastewater management and sustainable agriculture. They also covered horizontal issues, such as non-tariff measures, regulatory cooperation, trade facilitation and identification of EGS. While the work is ongoing, key insights so far are outlined in this document.[19]

In particular, as part of the discussions Members have identified: (i) indicative lists of EGS; (ii) trade barriers and supply chain bottlenecks; (iii) development perspectives; and (iv) opportunities and approaches for promoting and facilitating trade in EGS.

Indicative lists of Environmental Goods and Services

Members identified **indicative and non-exhaustive lists of goods and services** which can support achieving climate mitigation and adaptation goals in the sectors of: renewable energy, water supply and wastewater management, and sustainable agriculture. Examples of goods include photovoltaic cells for solar energy, gearboxes for wind turbines, generators for hydropower, electrolysers for green hydrogen production, and water drippers for efficient irrigation. Examples of services include certain engineering, testing and analysis, environmental consulting, operation, maintenance and repair, and recycling services.

Member experiences with lists of environmental goods, *inter alia*, include the Agreement on Climate Change, Trade and Sustainability (ACCTS; signed in 2024, expected entry into force 2026), the APEC list of environmental goods (54 goods; 2012), the EU-New Zealand FTA (48 goods; 2023), the Singapore-Australia Green Economy Agreement (SAGEA; 372 goods; 2022), and the New Zealand-UK FTA (293 goods; 2022). Analytical lists, *inter alia*, include the OECD's Combined List on Environmental Goods (CLEG; 345 goods). Experiences regarding environmental services include the ACCTS, EU-New Zealand FTA, SAGEA, and the 2021 APEC reference list of environmental and environmentally related services.

Several approaches and elements that might be considered in the **identification of EGS** were discussed and shared through Members experiences. These approaches included considering the environmental objective, end-use, and/or environmental impacts along the lifecycle, and using ex-outs where relevant.

Use of EGS lists: Experiences shared by Members highlighted that an approach of non-binding indicative lists of goods and services can be a useful tool and guide discussions on cooperation in areas such as technical standards and domestic regulatory regimes. Members also identified the use of indicative lists to be useful for guiding discussions on cooperation in areas such as technical standards and domestic regulatory regimes, and in signalling policy intent to markets and businesses even where liberalization did not occur. To encourage trade and investment in EGS, trade and economic cooperation agreements include different provisions, notably aiming at strengthening

[19] See the Analytical Summary (WT/MIN(26)/22/Add.3) for the extensive depiction of TESSD discussions on EGS.

bilateral and international cooperation, addressing tariffs and non-tariff barriers, or setting up mechanisms for review and update of EGS lists.

Development considerations

Trade in EGS can bring significant benefits to developing Members. Trade in EGS support access, affordability and deployment of climate-friendly technologies goods and services, which can support developing economies be integral part of sustainable and inclusive value chains on renewable energy, water management and sustainable agriculture sectors. It could help with fostering competitiveness, investment and economic development, while supporting sustainable development and climate mitigation and adaptation. Opportunities also exist for export of EGS and participation in value chains for renewable energy and sustainable agriculture.

Trade in EGS can complement domestic policy and international collaboration efforts in supporting developing economies overcome some **challenges they face, including**: access to finance and investment for climate change mitigation and adaptation; restricted market access; weak or inadequate infrastructure (e.g. logistical constraints in transport, electrical grids); limited access to technology and shortage of specialist know-how and skilled workers; complex or unclear regulatory environments and limited institutional capacity and lack of robust quality infrastructure – which includes facilities to develop, test and certify standards, technical regulations, and conformity assessment procedures. These challenges could be addressed, *inter alia*, by: improving access to international finance; facilitating to access to environmental goods; strengthening international cooperation through technical assistance and capacity building to help develop a skilled local workforce and invest in robust quality infrastructure; and further including developing countries' perspectives in the development of international standards also helps incorporate their perspectives into the global framework.

Developing economies can further **seize opportunities for inclusive and sustainable growth** through active participation in EGS value chains. Opportunities for local businesses and employment exist through fostering local value creation on ancillary components and services. Many developing economies are rich in natural resources and solutions, and could leverage comparative advantages and help diversify economies in sectors such as green hydrogen production, biofuels or sustainable agriculture. Finally, regional integration can help create larger markets, providing the necessary scale for new sustainable industries to become competitive, and strengthen the position of developing Members in global trade.

Opportunities and approaches to promote and facilitate trade in EGS

Members sought to identify opportunities and approaches that they have taken, or consider helpful, to promote and facilitate trade in environmental goods and services. While exchanges have taken place mainly in a sector-specific context, a number of opportunities and approaches identified tend to be more horizontal in nature, and others can be adapted to different sectors.

The opportunities and approaches to promote and facilitate trade in EGS discussed in TESSD meetings included: **enhancing transparency and knowledge sharing** which would increase access to information on regulatory frameworks and trade barriers and could lower trading costs, including for MSMEs; **providing technical assistance, capacity building and technology transfer on voluntary and mutually agreed terms** which could support greater uptake and deployment of EGS; **promoting of public-private partnerships** to drive investment and innovative solutions; **good regulatory practices and regulatory cooperation** could reduce the time and costs required for dealing with administrative procedures and might be particularly beneficial for MSMEs; and **market access, trade facilitation and preferential treatment** for EGS could improve uptake and deployment, helping reflect positive environmental externalities.

ANNEX IV: CIRCULAR ECONOMY-CIRCULARITY WG OUTCOME DOCUMENT – OVERVIEW

The Circular Economy-Circularity Working Group outcome document (WT/MIN(26)/22/Add.4) provides a compilation of trade aspects and practices for circular economy in four sectors: textiles; renewable energy; batteries; and electronics. The document aims to describe how trade and trade measures can support enhancing circularity. Building upon a wealth of knowledge on circular economy in the sectors, it presents trade aspects and their linkages and impacts on circularity, along the sectoral value chains. It further highlights Members' examples and practices. It is a non-binding document, and the considerations and Member practices included therein are without prejudice to Members' rights and obligations under the WTO Agreements.

Below, the 'Overview' section of the outcome document is reproduced. More information and concrete examples of policies shared by Members are available in the document.

OVERVIEW

The concept of a circular economy seeks to maximize the value of goods and materials and to keep them in the economy as long as possible through strategies such as reduction, reuse, and recycling. By doing so, it supports economic growth, resource efficiency and conservation, as well as sustainability.

Trade and trade policies play a crucial role in accelerating the transition to a more circular economy. Trade enables resource efficiency by connecting markets, extending the lifespan of products and materials, and reducing the demand for new resources. It does so by helping retain value within the value chain through reuse, repair, remanufacturing, disassembly, and recycling across borders. Integrating circular economy principles into international trade policies can foster more sustainable global trade, delivering economic, social, and environmental benefits for all.

Circular economy is not limited to actions taken at the end of life of a product, such as recycling; it encompasses the entire lifecycle of a product, including through reduction, reuse, and product life extension.

Each stage of this lifecycle can have a trade element. Therefore, key actions to promote circular economy, such as designing for circularity – prioritizing material efficiency and the use of secondary raw materials – and improving capacities for repair, reuse, and recycling, could benefit from taking the cross-border trade perspective into account. Effective communication and collaboration among manufacturers, repairers, and recyclers, including those situated in different jurisdictions, are vital to fostering circularity throughout the global value chain.

In exploring best practices and opportunities for voluntary actions and partnerships, discussions in the TESSD Working Group on Circular Economy have identified several recurring themes and opportunities. These are relevant across sectors such as textiles, batteries, renewable energy, and electronics, as well as in broader, crosscutting contexts. This high-level overview summarizes the key horizontal insights obtained from the analysis of these four sectors.

Transparency

Transparency is fundamental to circular economy, enabling trust on materials and between partners, traceability of goods and materials that are used in a circular system, and informed choices along the value chain.

The trade community can support efforts to better reflect circularity in product classifications and customs tools, such as in the Harmonized System (HS) codes, by establishing criteria to distinguish whether goods are being traded for reuse, refurbishment, remanufacturing, and recycling.

- Amongst issues discussed, for example, in the textiles sector, HS codes could be expanded to reflect not just used textiles, considered textile waste, but also the quality, durability, recyclability and reusability of textile products, while developing solutions for customs officials to determine these by visual inspection.

Trade policy can also promote transparency and traceability through labelling, product passports, or other systems that provide circularity-relevant information. Ensuring that products do not contain substances or materials that hinder safe circularity is particularly important. As demand for recycled content grows, tracking and ensuring the quality and safety of recycled materials will help ensure trust on recycled content and divert end-of-life products from landfills thus preventing further environmental pollution.

- In the batteries sector, battery passports offer a digital solution, providing standardized, verifiable data on a battery's origin, composition, carbon footprint, and compliance with due diligence commitments, such as eliminating child labour. This transparency supports responsible and sustainable production and consumption.

Standards and regulations

Standards and regulations, in line with the TBT agreement, are essential for an effective circular economy. Standards and regulations that set requirements for sustainable design can provide a common framework, ensure that products are designed for circularity, following the hierarchy of reducing, reusing, recycling and requiring features like easy disassembly and repairability. Working towards interoperability of regulations could enhance their effectiveness.

International cooperation is needed to ensure interoperability and alignment of standards and regulations. Trade policies should encourage the movement of circular products and services across borders. Inclusive, open dialogue among governments, industry, and civil society is crucial for interoperability of standards for circular economy and sharing best practices. Principles set out under the WTO Agreement on TBT are important when developing standards, regulatory practices, or mutual recognition in the context of circular economy.

International standards such as the ISO 59004 on "Circular Economy – Vocabulary, principles and guidance for implementation", which is designed to foster a shift towards a circular economy, with comprehensive guidance applicable to any type of organization, can support domestic efforts. In ISO 59004, principles of circular economy are presented, focusing on systemic thinking and the sustainable management of natural resources with the aim of generating environmental and socio-economic benefits. Another relevant example may be ASTM's new standard guide for principles of circular product design (E3461). Extended Producer Responsibility (EPR) is a policy approach that holds producers accountable for their products throughout their lifecycle, including at post-consumer stage.[20] EPR is considered a major instrument in support of the implementation of the waste management hierarchy.[21] and is meant to ensure that the polluter bears the cost of its pollution. Indeed, EPR represents an opportunity to finance the collection, sorting and recycling and environmentally sound disposal of end-of-life products, influence design, and improve the efficiency of materials, minimizing waste and incorporating end-of-life cost[22], further advancing a more circular economy. International cooperation can enhance its effectiveness where Members choose to adopt it.

- Over 80 countries have adopted EPR for e-waste as of 2024.[23] Nowadays, less than a quarter of electronic waste is properly collected and recycled, posing significant risks to health and the environment.

Trade facilitation

International collaboration towards trade facilitation would include simplified and efficient procedures for circular economy products, and deliver economic benefits, provided they are implemented within regulatory frameworks that protect human health and the environment. Cooperation with customs authorities on identifying circular products and their intended use is essential.

- Recent regional or bilateral trade agreements have included Circular Economy as a chapter or a topic of their trade agreements, often with a focus on renewable energy.

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[20] OECD (2024), Extended Producer Responsibility: Basic facts and key principles.
[21] BRS Convention (2019), Practical manual on Extended Producer Responsibility, adopted by COP14, UNEP/CHW.14/5/Add.1.
[22] Circular Innovation Lab presentation to TESSD WG on Circular Economy (Sept. 2023), Study on items shipped for reuse and EPR fees.
[23] Global E-waste Monitor 2024: GEM_2024_18-03_web_page_per_page_web.pdf.

Waste management

Environmentally sound management of waste is enabled by product design, transparency and traceability. in this context, facilitating the movement of secondary materials, recycled goods, and circular services including on waste management, can both drive and benefit from improved waste management. Effective systems for collection, sorting, and classification of end-of-life products are critical, especially in developing countries. Trade can also support investments in environmentally sound waste management and services that extend product lifetimes. The Basel Convention's Prior Informed Consent (PIC) procedure for hazardous waste, a "notice and consent" regime, was referenced in the TESSD discussions as necessary to protect human health and the environment, and prevent illegal waste trade.

Capacity building and technical assistance

An inclusive circular economy ensures that all stakeholders benefit economically, without solely bearing the costs of pollution. Developing economies, in particular, can leverage international trade to access technologies and tools that enable circular activities. Capacity building and technical assistance are vital to scaling up a more circular economy globally and ensuring that developing countries can fully participate and benefit.

- For example, some international projects discussed in the meetings focus on supporting SMEs in developing economies, on producing goods with minimal materials, reusing, refurbishing, or recycling, while other projects focus on building capacity on standards infrastructures, customs procedures (e.g. on transparency, or to conduct internationally recognized inspections and testing and certification).

Technology and other trade-related aspects of cooperation

Environmentally sound technologies, goods and services may be helpful inputs, from manufacturing through end-of-life stages, supporting the production of more circular and sustainable sectors, addressing technology and infrastructure gaps. Trade-related cooperation could be international, as well as regional or bilateral. Bilateral or regional trade agreements focusing on sustainability can be a vehicle to establish trade-related cooperation in support of identified key sectors.

The present document provides an in-depth analysis of trade and circular economy, with a focus on the afore mentioned four sectors, based on discussions in TESSD. It helps illustrate, through specific examples in each of the sectors, trade aspects identified in this overview, and related Member practices on circular economy.

Circular Economy Plans, Roadmaps or Strategies

Many Members have developed Circular Economy plans or roadmaps, under the auspices of multiple ministries, which articulate visions or strategies across a wide range of sectors and can encompass different policy tools. Some examples shared include:

- EU: Adopted in March 2020, the second Circular Economy Action Plan is one of the main building blocks of the European Green Deal, Europe's agenda for sustainable growth. The action plan includes initiatives addressing the entire life cycle of products, including how products are designed, as well as ensuring that waste is prevented and that used resources are kept in the EU economy for as long as possible.

- Mauritius Circular Economy Roadmap and Action Plan (2023-2033), in line with the Government Programme 2025-2029 for advancing Mauritius as a green economy that encourages recycling and fosters a circular economy.

- Peru's Supreme Decree No. 003-2025-MINAM (published on 25 February 2025) approved the National Roadmap for Circular Economy to 2030. The National Circular Economy Roadmap (HRNEC) will seek by 2030 to generate enabling conditions to make the transition to a circular economy viable through four strategic objectives: (1) Circular governance and policies, (2) Innovation and circular business, (3) Sustainable consumption and circular culture, and (4) Circular territories and cities.

- The Swiss Federal Assembly adopted the Parliamentary Initiative 20.433 "Strengthening Switzerland's Circular Economy" on 15 March 2024. The revisions to the Environmental Protection Act, the Energy Act, and the Federal Act on Public Procurement establish an overarching legal framework to reinforce the circular economy in Switzerland.

- The Kingdom of Saudi Arabia's commitment to environmental sustainability, as outlined in Vision 2030 aligns with national and global goals, including reducing carbon emissions, achieving carbon neutrality and fostering a circular economy. The Kingdom of Saudi Arabia has implemented more than 30 Circular Carbon Economy initiatives across the energy system, enabling climate action while bolstering economic growth. The Circular Carbon Economy (CCE) framework is an integrated, inclusive and pragmatic approach to managing emissions.

Amongst others, Chatham House, with support from UNIDO, has conducted a global stocktake of national circular economy roadmaps, including 75 national roadmaps and 2882 individual policy actions.[24] In their analysis, they indicated that policies included in the roadmaps sometimes covered fiscal instruments such as taxes and duties, public financing or investment funds, as well as producers or products requirements.

Some regional platforms have also been focusing on Circular Economy, such as the ASEAN Framework on Circular Economy, African Circular Economy Roadmap or the Circular Economy Coalition of Latin America and the Caribbean's mission is to provide a regional platform to enhance inter-ministerial, multi-sectoral and multi-stakeholder cooperation to increase knowledge and understanding about the circular economy, facilitate training, training and technical assistance for the development of circular economy and sustainable consumption and production public policies.

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[24] Circulareconomy.earth and Chatham House, Global stocktake of national circular economy roadmaps, also presented in TESSD meeting in April 2024.